

Graph-Based Compression of DEM-Resolved Agglomerate Morphology

Ali Khalifa and Heiko Briesen

Chair of Process Systems Engineering, TUM School of Life Sciences,
Technical University of Munich, Freising, Germany

Abstract

The internal morphology of particle agglomerates can affect oxygen access beyond conventional properties such as particle number, size and nominal fractal dimension. In this preliminary study, 135 synthetic monodisperse agglomerates were generated, relaxed by the discrete-element method and converted into particle-contact graphs. The supervised target was the mean geometric exposure: for each particle, uniformly distributed outward surface rays were tested for obstruction by all other particles, and the unobstructed fraction was averaged over the aggregate.

A geometry-aware graph neural network (GNN) was evaluated using grouped five-fold cross-validation. Only the learned latent representation was supplied to the prediction head. The predefined compactness rule selected one latent variable, which predicted mean geometric exposure with RMSE 0.0160, MAE 0.0128 and $R^2 = 0.954$. The latent coordinate was most strongly associated with the requested fractal dimension and fractal prefactor ($|\rho| = 0.800$), and six conventional size and contact descriptors reconstructed it with grouped-CV $R^2 = 0.991$. Nevertheless, the GNN predicted exposure more accurately than particle count alone ($R^2 = 0.160$), the best individual classical descriptor model ($R^2 = 0.511$), and combined classical descriptor models (best $R^2 = 0.922$). The learned coordinate is therefore a compact task-specific representation of geometric exposure, although the target remains a line-of-sight proxy rather than a resolved oxygen-transport quantity.

Keywords: particle agglomerates; discrete-element method; contact graphs; graph neural networks; latent morphology representation; transport accessibility

1 Introduction

Transport models for biological and cohesive particle aggregates commonly describe morphology using a small set of integral quantities, such as aggregate diameter, porosity

or fractal dimension. Such descriptors are useful, but they do not uniquely determine the internal arrangement of the primary particles. Aggregates with similar integral properties can still differ in connectivity, surface exposure, shielding and the existence of internal transport pathways. These structural differences may subsequently influence substrate or oxygen accessibility and therefore the effective activity of an aggregate.

The broader research concept is to derive morphology-aware oxygen-transport closures from particle-resolved CFD–DEM simulations and use them in coarse-scale CFD–PBM or Euler–Lagrange reactor models. In that future framework, a graph neural network would encode the detailed particle structure into a compact morphology descriptor, while a reduced closure would combine this descriptor with local reactor conditions to predict oxygen uptake, dynamic response and the fraction of oxygen-limited cells.

Before conducting expensive coupled flow and transport simulations, the present DEM-only study tests the feasibility of the structural part of this concept. Its central hypothesis is that a particle-contact graph contains nonlocal morphological information that can be compressed into a low-dimensional latent representation without losing the information required to predict geometric surface exposure. The study therefore addresses three preliminary questions:

1. Can generated agglomerates be mechanically relaxed and converted consistently into particle-contact graphs?
2. Can a GNN predict mean geometric exposure for parameter combinations withheld from training?
3. How many latent variables are required to retain this structural information?

The calculated target is intentionally limited to a geometric line-of-sight proxy. This separation is important: the study validates the computational workflow and the concept of morphology compression, while the physical relation to oxygen transport remains a task for the proposed particle-resolved project.

The proof-of-concept study proceeds in six steps. First, synthetic agglomerates are generated with different particle counts and physical sizes, as well as with equal particle counts but different compactness settings and random particle arrangements. Second, the artificial structures are relaxed in a physical DEM environment to test whether they remain mechanically stable under the selected particle and contact forces. Third, geometric

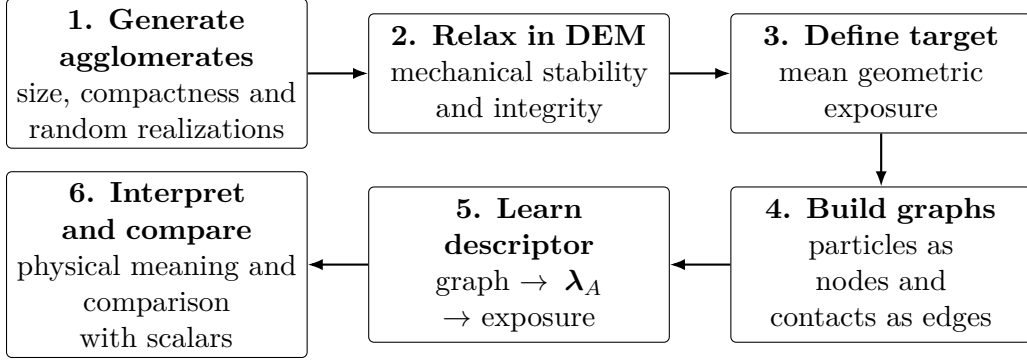


Figure 1. Six-step workflow used to generate and relax synthetic agglomerates, represent them as graphs, learn a compact structural descriptor and evaluate its physical meaning and predictive value.

exposure is calculated from unobstructed outward surface rays. Fourth, each relaxed agglomerate is converted into a particle-contact graph. Fifth, a GNN is trained to compress the graph into one or more latent descriptors while retaining the information needed to predict mean exposure. Finally, the learned descriptor is interpreted by comparing it with conventional physical and graph-based measures, and its predictive value is tested against particle count alone and against models using several conventional descriptors together. The complete workflow is summarized in Fig. 1.

2 Methodology and computational workflow

2.1 Synthetic agglomerate ensemble

The preliminary data set consists of monodisperse agglomerates generated with the tunable particle-cluster and cluster-cluster aggregation formulation of Filippov, Zurita and Rosner [1], following the efficient hierarchical implementation described by Skorupski et al. [2]. The method prescribes the mass-radius relation

$$N = k_f \left(\frac{R_g}{r_p} \right)^{D_f}, \quad (1)$$

where N is the number of primary particles, r_p is their radius, R_g is the radius of gyration including the finite primary-particle radius, D_f is the requested fractal dimension and k_f is the fractal prefactor.

Generation begins with stochastic 25-particle seed clusters. Starting from two touching

Table 1. Design variables used in the DEM investigation.

Design variable	Values
Primary-particle diameter d_p	1.0, 1.5, 2.0 μm
Number of particles N	50, 100, 200
Requested (D_f, k_f) and morphology	(1.8, 1.3) open; (2.2, 1.1) intermediate; (2.6, 0.8) comparatively compact
Random realizations	5 per parameter combination
Total number of cases	135

particles, further particles are placed at random orientations on the radial shell required by Eq. (1); placements are accepted only when they establish contact without overlap. Equal-sized seed clusters are subsequently combined hierarchically. Before each merge, both subclusters are randomly rotated. Their prescribed center-of-mass separation R_{12} follows from the parallel-axis relation,

$$R_{12}^2 = r_p^2 \frac{N^2}{N_1 N_2} \left(\frac{N}{k_f} \right)^{2/D_f} - \frac{N}{N_2} R_{g,1}^2 - \frac{N}{N_1} R_{g,2}^2, \quad N = N_1 + N_2. \quad (2)$$

A Monte Carlo orientation search identifies a non-overlapping configuration whose first-contact separation agrees with R_{12} within the prescribed tolerance. The clusters are placed at this analytically determined first-contact distance, guaranteeing connectivity without overlap while allowing only a small, recorded deviation from the requested mass–radius law. The particle counts $N = 50, 100$ and 200 correspond to one, two and three hierarchical merging levels, respectively, from the common 25-particle seeds.

Three morphology settings were used: $(D_f, k_f) = (1.8, 1.3)$ for open, $(2.2, 1.1)$ for intermediate and $(2.6, 0.8)$ for comparatively compact agglomerates. The word “compact” is relative to the other fractal cases: the cluster–cluster construction creates contact-connected fractal structures and does not imply dense, nearly spherical packing. Dense spherical clusters would require an additional restructuring or densification procedure [3].

The data set comprises three primary-particle diameters, three particle counts, three morphology settings and five independent random realizations per design point,

$$3 \text{ diameters} \times 3 \text{ particle counts} \times 3 \text{ morphologies} \times 5 \text{ realizations} = 135 \text{ agglomerates.}$$

The details are summarized in Table 1.

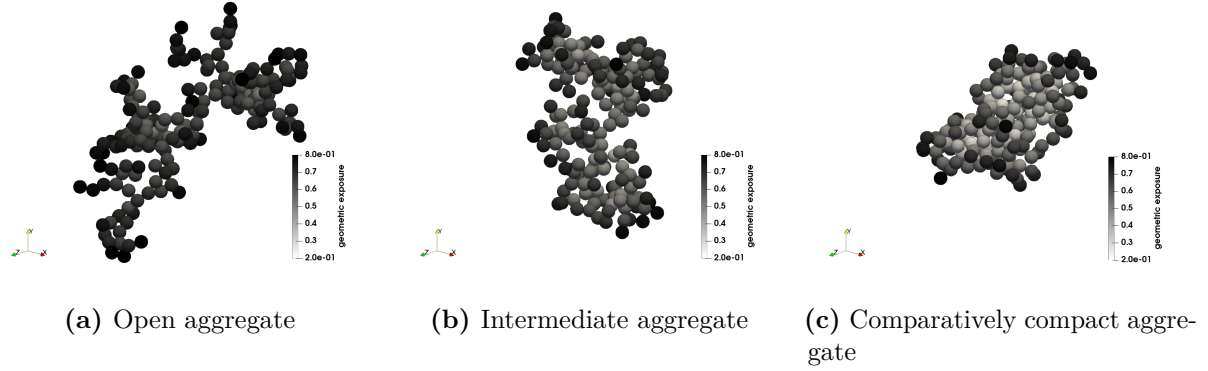


Figure 2. Representative post-DEM agglomerates illustrating the range of morphologies in the preliminary data set. The three agglomerates have the same primary-particle diameter ($d_p = 2 \mu\text{m}$) and particle count ($N = 200$) but different requested (D_f, k_f) morphology settings. The particles are coloured according to their particle-level geometric exposure a_i^{geo} , using a common colour scale from 0 (all sampled outward directions blocked) to 1 (all sampled outward directions unobstructed).

Each realization was assigned a unique random seed. The generated particle coordinates and radii were exported in formats suitable for DEM relaxation, graph construction and visualization. All particles represented the same material; no permanent bonds were prescribed, and the initial translational and angular velocities were zero.

2.2 DEM relaxation and stability assessment

The generated structures were mechanically relaxed using a soft-sphere DEM formulation. Normal contact was described by Hertzian elastic repulsion and regularized van der Waals attraction. The adhesive interaction is governed by the Hamaker constant A_H and a minimum surface separation $h_{\min}$, which removes the singularity of the idealized interaction at vanishing separation [4, 5]. Tangential resistance was evaluated from the contact history and limited by Coulomb friction. Rolling and torsional resistance were also included. No permanent bonds were introduced; the relaxed structures were maintained solely by adhesion and contact friction. The applied contact models are summarized in Table 2.

Although the numerical setup permits multiple material types, identical properties were used for every interaction in the present monodisperse study.

The primary-particle diameters of $1.0\text{--}2.0 \mu\text{m}$ represent the characteristic micrometre scale of many bacterial cells [6]. The range is intentionally narrow: its purpose in this preliminary study is to test whether the graph workflow remains consistent when

Table 2. Contact formulation used for DEM relaxation.

Model component	Physical formulation
Normal interaction	Hertzian elastic repulsion combined with regularized van der Waals attraction
Tangential interaction	History-dependent tangential force with Coulomb limitation
Rolling resistance	Directional rolling-resistance torque
Torsional resistance	Resistance to relative twisting at a contact
Near-contact regularization	Limitation of nonphysical normal-force states near contact
Permanent bonds	None

Table 3. Particle and contact parameters used in the DEM relaxation. The primary-particle diameter was varied, while the remaining properties were held constant.

Property	Symbol	Value
Particle shape	—	Sphere
Primary-particle diameter	d_p	1.0, 1.5, 2.0 μm
Reference particle density	ρ_p	1050 kg m^{-3}
Young’s modulus	E	$1.0 \times 10^9 \text{ Pa}$
Poisson ratio	ν	0.45
Coefficient of restitution	e	0.20
Sliding-friction coefficient	μ	0.30
Rolling-friction coefficient	μ_r	0.01
Hamaker constant	A_H	$3.028 \times 10^{-20} \text{ J}$
Minimum separation distance	$h_{\min}$	$4.0 \times 10^{-10} \text{ m}$

the physical length scale changes, while particle number and morphology are varied independently. The density $\rho_p = 1050 \text{ kg m}^{-3}$ represents a water-rich particle only slightly denser than the surrounding liquid. Measurements for *Escherichia coli*, for example, place the buoyant density near $1080\text{--}1105 \text{ kg m}^{-3}$ [7, 8]. The slightly lower reference value used here is therefore an idealized near-neutral-buoyancy choice rather than a species-specific measurement. The Poisson ratio $\nu = 0.45$ represents a nearly incompressible effective particle.

The remaining mechanical parameters define a controlled relaxation model and must not be interpreted as a calibrated constitutive description of a particular microorganism. Reported effective bacterial-envelope moduli are commonly in the MPa range and depend strongly on species, physiological state and measurement protocol [9, 10]; the present value $E = 1 \text{ GPa}$ was selected as a stiff numerical contact parameter to restrict overlap during relaxation. The low restitution coefficient provides strong damping, whereas the sliding- and rolling-friction coefficients resist relative translation and rotation. The adopted Hamaker constant is of the conventional 10^{-20} J order associated with dispersion interactions, but its effective value depends on the interacting materials and the intervening liquid [5]. Accordingly, elasticity, dissipation, friction and adhesion should be varied in a sensitivity study and calibrated before applying the model to a specific bioaggregate. The relevant particle and contact parameters used in the present DEM relaxation simulations are listed in Table 3.

The relaxation simulations were performed with a physical time step of $\Delta t = 5.0 \times 10^{-11} \text{ s}$ and were run for up to 300000 time steps, corresponding to a maximum physical duration of $1.5 \times 10^{-5} \text{ s} = 15 \mu\text{s}$. The purpose of this DEM stage was to obtain mechanically stable structures and to verify that the artificially generated agglomerates remained intact under the selected elasticity, dissipation, friction and adhesive-contact conditions without permanent bonds. The suitability of the time step was checked every 1000 steps against the characteristic contact and neighbour-list time scales. Particle and pair-interaction data were written every 20000 steps, corresponding to output intervals of $1 \mu\text{s}$. Gravity was not activated because the simulation focused on internal contact-scale relaxation without sedimentation or an externally imposed loading direction.

The particle output contained positions, radii, particle IDs and velocities, whereas the pair output contained interacting particle IDs, gap, overlap, contact area, normal

and tangential forces, total force and torque. These data were converted to VTK for inspection and subsequent graph construction. Because the adhesive interaction also acts across a finite gap, a reported particle pair does not necessarily represent a geometrically touching contact. The pairs were therefore classified using

$$\delta_{ij} = r_i + r_j - \|\mathbf{x}_i - \mathbf{x}_j\|. \quad (3)$$

Pairs satisfying $\delta_{ij}/d_{\text{ref}} \geq -10^{-6}$ were treated as touching. This small tolerance retains numerically near-touching pairs while excluding separated neighbours interacting only through finite-range adhesive forces.

For each case, particle and contact VTK files were matched by time step. An agglomerate was considered relaxed when, over three consecutive output intervals,

$$\frac{\Delta x_{\text{RMS}}}{d_p} \leq 10^{-4}, \quad 1 - \frac{|C_k \cap C_{k+1}|}{|C_k \cup C_{k+1}|} \leq 0.01, \quad (4)$$

where C_k denotes the set of particle contacts at time step k . These criteria require both the particle positions and the contact network to become nearly stationary.

Agglomerate integrity was evaluated independently using connected-component analysis. Each particle was treated as a graph node and every reported particle-pair interaction as an undirected edge. All 135 final structures contained exactly one connected component with a largest-component fraction of unity. Thus, none of the artificially generated agglomerates fragmented during the DEM relaxation.

2.3 Effective fractal-dimension characterization

The fractal dimensions specified during agglomerate generation define the requested morphology settings. Since the generated structures contain only 50–200 primary particles and are subsequently relaxed by DEM, these input values should not be interpreted as exact measured properties of every final structure. The relaxed agglomerates were therefore characterized using two complementary finite-scale methods. Both methods are commonly used to characterize particle aggregates, although their estimates can depend on the available scale range and the finite number of primary particles [11, 12].

The first estimate was based on the mass–radius scaling relation commonly used for

fractal particle aggregates [13, 11],

$$N(r) \propto r^{D_f}, \quad (5)$$

where $N(r)$ is the number of particle centres within a distance r from the agglomerate centre of mass. After calculating the distance of every particle centre from the agglomerate centre of mass, the particles were sorted according to this radial distance. The effective fractal dimension was then obtained from the slope of

$$\log N(r) = D_f \log r + C. \quad (6)$$

Only the range containing 10–80 % of the particles was included in the regression. The innermost region was excluded because it contains only a small number of particles, while the outermost region is strongly affected by the finite boundary of the agglomerate.

A second estimate was obtained using the box-counting method [14, 15, 12]. The particle centres were covered by a three-dimensional Cartesian grid with box size ϵ , and the number of boxes containing at least one particle centre, $N_{\text{box}}(\epsilon)$, was counted. The effective box-counting dimension follows from

$$N_{\text{box}}(\epsilon) \propto \epsilon^{-D_f}, \quad (7)$$

and was calculated as the slope of $\log N_{\text{box}}$ against $\log(1/\epsilon)$.

Before box counting, each agglomerate was translated to its centre and aligned with its principal axes. Twelve logarithmically spaced box sizes were evaluated. For each box size, 32 shifted grids were tested and the smallest occupied-box count was retained to reduce sensitivity to the arbitrary grid origin. Scales for which nearly all particles occupied separate boxes, as well as scales for which the complete agglomerate occupied approximately one box, were excluded. Such scale selection is important because both limits contain little information about the intermediate spatial scaling [15, 12].

2.4 Graph representation and geometric-exposure target

Each selected agglomerate snapshot was represented as a contact graph. Primary particles form the nodes, while only geometrically touching or numerically near-touching pairs

satisfying $\delta_{ij}/d_{\text{ref}} \geq -10^{-6}$ form the edges. Separated attractive neighbours were excluded. This choice gives the graph a clear geometrical interpretation and prevents its topology from depending on the finite interaction cutoff of the adhesive model. Every retained pair was stored in both directions for message passing. Thus, an agglomerate with N particles and N_c retained contacts has N nodes and $2N_c$ directed edges. All 135 agglomerates yielded valid graphs and were included in the analysis; no case was skipped.

Three node features were available after preprocessing. The minimal model used the two nonconstant structural scalars

$$\mathbf{x}_i = \left[\frac{|\mathbf{x}_i - \mathbf{x}_c|}{R_g}, z_i \right], \quad (8)$$

where d_{ref} is twice the median particle radius, $\mathbf{x}_c$ is the agglomerate center, R_g is the radius of gyration, and z_i is the coordination number. The third available node feature, r_i/d_{ref} , is approximately 0.5 for all present monodisperse particles and was therefore omitted from the minimal model.

The processed DEM data contained 22 available contact features, including normalized distance and overlap, gap, contact area, force and torque quantities and contact-state flags. For the minimal model, only the normalized centre distance d_{ij}/d_{ref} was selected from these stored contact features, because connectivity and interparticle geometry are the quantities directly relevant to the geometric-exposure target considered here. The graph connectivity and normalized relative-position vector were additionally used for message passing. The remaining available contact features were retained for possible sensitivity studies but were not supplied to the reported model.

Geometric exposure was calculated independently of the convex hull. For each particle i , $M = 512$ approximately uniform unit directions $\mathbf{s}_m$ were generated on the sphere using a deterministic Fibonacci construction. A ray was started immediately outside the particle surface at $\mathbf{x}_i + (r_i + \varepsilon)\mathbf{s}_m$ and propagated along the outward normal $\mathbf{s}_m$. As illustrated in Fig. 3, a direction was classified as blocked if the corresponding ray intersected any other particle; otherwise, it was classified as unobstructed. Let $b_{im} = 1$ for a blocked ray and

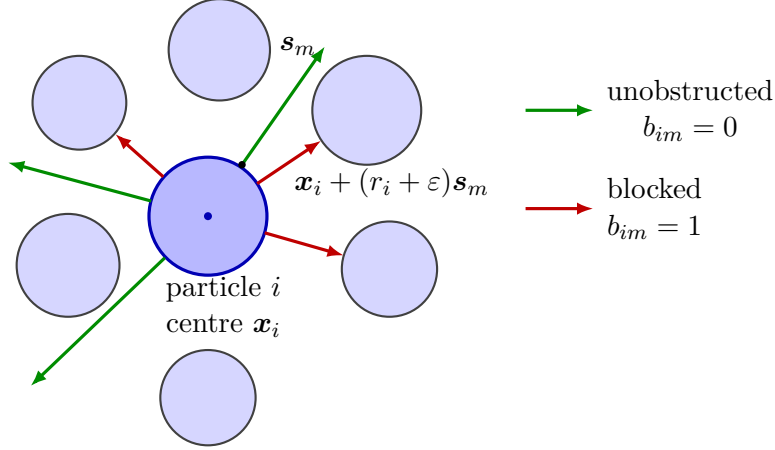


Figure 3. Schematic of the ray-tracing calculation for one particle i . Rays start just outside its surface and follow sampled outward directions $\mathbf{s}_m$. A ray is blocked when it intersects another particle; otherwise, it contributes to the exposed fraction. For clarity, only a two-dimensional section and a small subset of rays are shown; the actual calculation distributes $M = 512$ directions over the complete sphere.

$b_{im} = 0$ for an unobstructed ray. The particle-level exposure and the aggregate target are

$$a_i^{\text{geo}} = \frac{1}{M} \sum_{m=1}^M (1 - b_{im}), \quad (9)$$

$$\bar{a}_{\text{geo}} = \frac{1}{N} \sum_{i=1}^N a_i^{\text{geo}}. \quad (10)$$

Thus, $a_i^{\text{geo}} = 1$ means that all sampled outward directions are unobstructed, whereas $a_i^{\text{geo}} = 0$ means that every sampled direction is blocked. The reported calculation used zero probe-radius inflation, so rays were blocked by the physical particle radii. Unlike the former contact-depth proxy, this definition can assign low or high exposure to particles that are close to the aggregate boundary but are not convex-hull vertices. It also accounts directly for shadowing by particles located farther along a surface direction. Nevertheless, $\bar{a}_{\text{geo}}$ remains a purely geometrical line-of-sight measure; it does not include pore diffusion, flow, reaction or oxygen uptake.

2.5 GNN compression and training

In addition to the stored node and edge features, the model uses the particle coordinates relative to the aggregate centre and normalized by the reference diameter d_{ref} . The corresponding relative-position vector is assigned to every directed edge. Random rigid rotations are applied during training to prevent the model from associating the target

quantity with a particular aggregate orientation.

The node and edge inputs are embedded into a hidden space of dimension 64. Six residual GINE message-passing layers with layer normalization and dropout are then applied. The message-passing depth allows information to propagate over extended paths through the aggregate, while the residual connections reduce optimization difficulties and the loss of local information. Graph-level mean, maximum and size-sensitive normalized-sum pooling are combined to form a graph-derived aggregate representation. This representation alone is compressed through the latent bottleneck into a morphology vector

$$\boldsymbol{\lambda}_A = [\lambda_1, \dots, \lambda_{n_\lambda}] \in \mathbb{R}^{n_\lambda}, \quad (11)$$

where the latent dimension n_λ is treated as a model hyperparameter. Candidate values $n_\lambda \in \{1, 2, 3, 4\}$ are evaluated by grouped cross-validation. Thus, the GNN learns the values of the latent components, whereas their required number is determined from the predictive performance and model-complexity comparison.

No separately calculated aggregate-level descriptors are supplied to either the latent bottleneck or the prediction head. Aggregate size, spatial extent and connectivity can nevertheless be inferred from the number and arrangement of nodes, their normalized coordinates, the contact topology and the size-sensitive normalized-sum pooling term. A two-layer prediction head maps only $\boldsymbol{\lambda}_A$ to the supervised exposure target,

$$\hat{y}_A = \hat{a}_{\text{geo}}. \quad (12)$$

Training used Adam with a learning rate of 10^{-3} , weight decay of 10^{-5} , batch size 16 and random seed 7. The targets were standardized using only the respective training data. A validation subset and early stopping were used to retain the best model in each fold instead of automatically selecting the last training epoch.

The assessment used grouped five-fold cross-validation. All five random realizations with the same (d_p, N, D_f, k_f) combination were assigned to the same test fold. This produced 27 parameter groups. Consequently, each reported prediction was produced by a model that had not seen another realization of the corresponding parameter combination during training. The predictions from the five held-out folds were combined to calculate

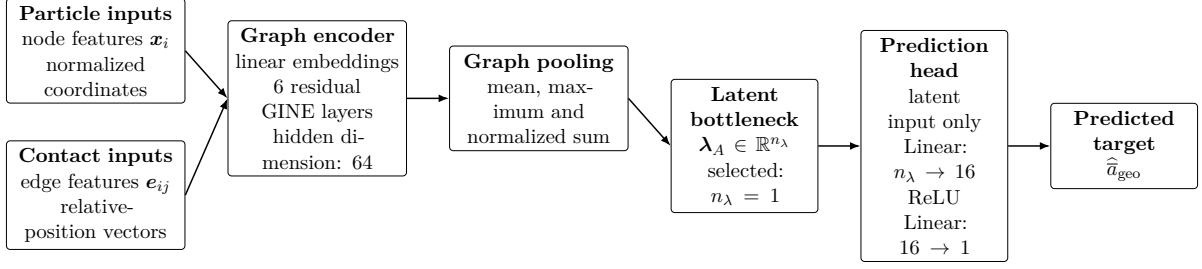


Figure 4. Architecture of the graph-based morphology model. Particle and contact information is processed by the message-passing graph encoder. The pooled graph representation is compressed into the latent morphology vector λ_A . A two-layer multilayer perceptron maps only this latent vector to mean geometric exposure. All components are trained jointly.

the out-of-fold metrics

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{j=1}^n (y_j - \hat{y}_j)^2}, \quad (13)$$

$$\text{MAE} = \frac{1}{n} \sum_{j=1}^n |y_j - \hat{y}_j|, \quad (14)$$

$$R^2 = 1 - \frac{\sum_{j=1}^n (y_j - \hat{y}_j)^2}{\sum_{j=1}^n (y_j - \bar{y})^2}. \quad (15)$$

Here, RMSE and MAE quantify the prediction error, whereas R^2 measures the fraction of the variation among the 135 aggregates explained by the model.

3 Results and interpretation

3.1 Latent dimension and predictive performance

Latent dimensions $n_\lambda \in \{1, 2, 3, 4\}$ were evaluated using identical grouped five-fold splits. The highest out-of-fold performance was obtained with four components ($R^2 = 0.969$), whereas the one-dimensional model reached $R^2 = 0.954$. The predefined selection rule retains the smallest dimension within 0.02 of the maximum. Since $0.954 > 0.969 - 0.02$, the selected representation is

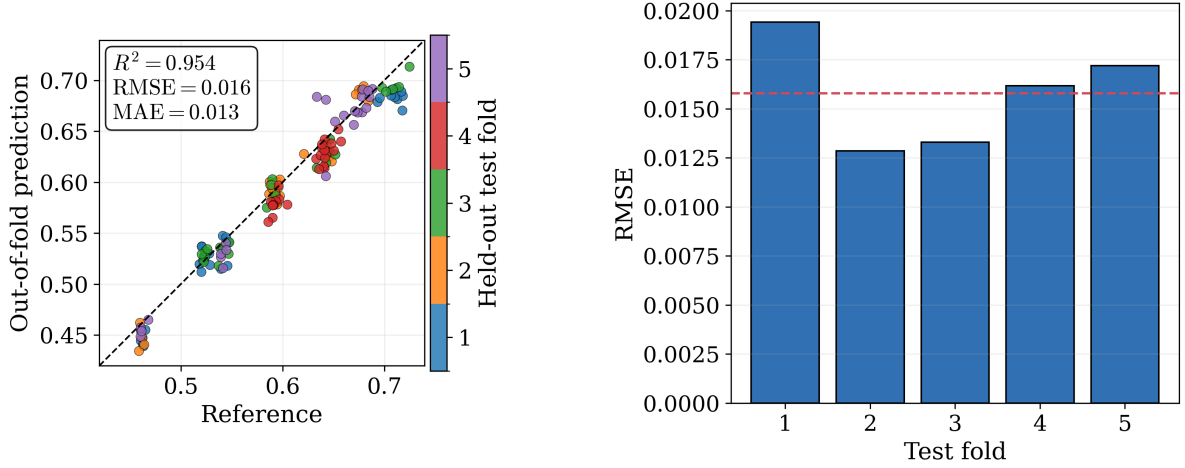
$$\lambda_A = [\lambda_1]. \quad (16)$$

This choice sacrifices only 0.0148 in R^2 relative to four components while providing a single interpretable morphology coordinate.

For the selected one-dimensional model, the combined out-of-fold RMSE was 0.0160, the MAE was 0.0128 and $R^2 = 0.954$. The five fold-specific RMSE values were 0.0194,

Table 4. Grouped out-of-fold performance for the candidate latent dimensions. The selected compact representation is highlighted in bold.

n_λ	$R^2_{a_{\text{geo}}}$	RMSE	MAE
1	0.954	0.0160	0.0128
2	0.966	0.0139	0.0113
3	0.958	0.0153	0.0119
4	0.969	0.0132	0.0106



(a) Reference particle-averaged geometric exposure versus its grouped out-of-fold GNN prediction. Colours identify the held-out test fold. (b) RMSE in each held-out test fold. The dashed line denotes the mean fold RMSE.

Figure 5. Prediction of mean geometric exposure with the selected one-dimensional GNN under grouped five-fold cross-validation: (a) out-of-fold predictions for all 135 agglomerates and (b) variation of prediction error among the held-out folds.

0.0128, 0.0133, 0.0162 and 0.0172. Performance also remained high when evaluated separately within the three particle-count classes: $R^2 = 0.856$, 0.975 and 0.944 for $N = 50$, 100 and 200, respectively. The lower value for $N = 50$ indicates that this subset remains the most difficult, but the model still captures morphology differences at fixed particle count.

3.2 Interpretation of the learned coordinate

After selecting $n_\lambda = 1$, final models were trained on all 135 graphs using random seeds 7, 17 and 27. Their coordinates were standardized, aligned in sign and averaged. This all-data coordinate is used only for descriptive interpretation and not as a leakage-free prediction feature.

Table 5 compares λ_1 with conventional descriptors. The strongest monotonic asso-

Table 5. Association of the exposure-specific latent coordinate with conventional descriptors, ordered by absolute Spearman correlation.

Descriptor	Pearson r	Spearman ρ
Requested fractal dimension D_f	-0.789	-0.800
Fractal prefactor k_f	0.778	0.800
Number of contacts N_c	-0.579	-0.596
Particle count N	-0.576	-0.544
Maximum coordination $z_{\max}$	-0.555	-0.541
Mean coordination $\bar{z}$	-0.532	-0.518
Contact density ρ_c	0.562	0.442
Radius of gyration R_g/d_{ref}	0.141	0.051
Primary-particle diameter d_p	-0.003	0.000

ciations were obtained with the requested fractal dimension ($\rho = -0.800$) and fractal prefactor ($\rho = 0.800$). These two effects cannot be separated in the present design because the three morphology classes pair D_f and k_f one-to-one. Contact count, particle count and the coordination measures showed moderate negative associations. In contrast to the former three-target latent coordinate, R_g/d_{ref} was essentially uncorrelated with the exposure-specific coordinate. The arbitrary sign of a one-dimensional latent variable should be kept in mind: reversing the sign would reverse all reported correlations without changing the model.

The five realizations in each of the 27 parameter groups had a mean within-group standard deviation of 0.0586 in the aligned coordinate, with group values ranging from 0.0106 to 0.123. Thus, stochastic variation within a design point was small relative to the unit standard deviation of the complete aligned coordinate.

Grouped-CV regressions were then used to reconstruct λ_1 from the available interpretation variables: d_p , requested D_f , k_f , N , R_g/d_{ref} , N_c , ρ_c , $\bar{z}$, $z_{\max}$ and the one-hot-encoded morphology class. The combined random forest reached $R^2 = 0.991$ and the combined linear model $R^2 = 0.988$. The best individual variables were requested D_f ($R^2 = 0.557$) and k_f ($R^2 = 0.536$); no single measured size or contact descriptor reconstructed the latent coordinate well. These results indicate that λ_1 is a compact combination of several morphology attributes, not a renamed version of particle count or radius of gyration. Because the all-data GNN generated the interpreted latent values before the reconstruction cross-validation, this analysis is descriptive rather than a fully nested independent validation.

Table 6. Grouped-CV reconstruction of the exposure-specific latent coordinate from conventional descriptors.

Reconstruction model	R^2
Combined random forest	0.991
Combined linear model	0.988
Linear: requested D_f	0.557
Linear: fractal prefactor k_f	0.536
Linear: maximum coordination	0.262
Linear: mean coordination	0.254
Linear: contact density	0.250
Linear: number of contacts	0.243
Linear: particle count	0.238
Linear: R_g/d_{ref}	-0.016
Linear: particle diameter	-0.066

3.3 Comparison with classical descriptors

The leakage-free target comparison reused exactly the grouped test folds of the GNN. The classical models received either one calculated descriptor or the following six descriptors simultaneously:

$$N, \quad R_g/d_{\text{ref}}, \quad N_c, \quad \rho_c, \quad \bar{z}, \quad z_{\text{max}}. \quad (17)$$

The target $\bar{a}_{\text{geo}}$ was never supplied as an input. A separate baseline received only the paired requested generation parameters D_f and k_f , without any descriptors calculated from the relaxed agglomerates.

The pure-graph GNN achieved the highest predictive performance, with $R^2 = 0.954$. The linear model combining the six calculated descriptors reached $R^2 = 0.922$, whereas the model based only on the requested values of D_f and k_f reached $R^2 = 0.604$. Thus, the prescribed morphology class contained substantial information about geometric exposure but did not reproduce the realization-, size- and relaxation-dependent variation captured by the particle graph. Because only the three paired combinations $(D_f, k_f) = (1.8, 1.3)$, $(2.2, 1.1)$ and $(2.6, 0.8)$ were investigated, the individual effects of D_f and k_f cannot be separated using the present dataset.

No individual calculated descriptor approached the performance of the combined models. The strongest individual result was obtained using a random-forest model of R_g/d_{ref} , which reached $R^2 = 0.511$. Linear maximum coordination reached $R^2 = 0.296$, whereas the best particle-count model reached only $R^2 = 0.160$. The detailed graph

Table 7. Grouped out-of-fold prediction of mean geometric exposure. The D_f - k_f baseline uses only the paired requested generation parameters. For each individual calculated descriptor and for the D_f - k_f baseline, the better of the linear and random-forest models is shown.

Model or input descriptor	RMSE	MAE	R^2
Pure-graph GNN	0.0160	0.0128	0.954
Six descriptors, linear	0.0209	0.0183	0.922
Six descriptors, random forest	0.0235	0.0168	0.902
Requested D_f and k_f , linear	0.0472	0.0399	0.604
R_g/d_{ref} , random forest	0.0524	0.0383	0.511
Maximum coordination, linear	0.0629	0.0516	0.296
Mean coordination, linear	0.0670	0.0577	0.201
Number of contacts, linear	0.0685	0.0580	0.164
Particle count, linear	0.0687	0.0582	0.160
Contact density, linear	0.0701	0.0603	0.123

therefore contains predictive information about directional surface obstruction that is not captured by any individual conventional scalar and is not completely recovered by their six-variable combination.

4 Conclusions and next steps

This study establishes an exposure-specific DEM-to-graph-to-GNN workflow for 135 relaxed agglomerates. Geometric exposure is defined by outward-normal line-of-sight tests over the complete particle surfaces and therefore avoids classifying accessibility solely through convex-hull membership or contact depth. The selected one-dimensional GNN predicted this target with grouped out-of-fold $R^2 = 0.954$, RMSE 0.0160 and MAE 0.0128.

The learned coordinate is task-specific and combines several morphology attributes. It was most strongly associated with the paired generation parameters D_f and k_f , while particle count and contact-network measures showed moderate correlations. Conventional descriptors reconstructed the all-data latent coordinate accurately when combined, but this descriptive result does not replace leakage-free target evaluation. In the direct target comparison, the GNN outperformed the best combined classical model ($R^2 = 0.922$), the paired requested- D_f - k_f baseline ($R^2 = 0.604$), the best individual descriptor model ($R^2 = 0.511$), and particle count alone ($R^2 = 0.160$). Thus, the generation setting explains a substantial part of the exposure variation, but the detailed relaxed particle graph is needed for the much higher predictive accuracy obtained here.

The target is nevertheless a geometric proxy. It treats all unobstructed rays equally and does not account for tortuous diffusion, pore bottlenecks, fluid motion, oxygen consumption or reaction–transport coupling. The next stage should examine ray-number convergence and probe-radius sensitivity, expand the morphology ensemble, and compare $\bar{a}_{\text{geo}}$ with oxygen uptake, response time and oxygen-limited particle fractions obtained from resolved transport calculations. That comparison will establish whether the present exposure coordinate is physically useful or must be supplemented by transport-aware graph information.

References

- [1] A. V. Filippov, M. Zurita, and D. E. Rosner. “Fractal-like aggregates: Relation between morphology and physical properties”. In: *Journal of Colloid and Interface Science* 229 (2000), pp. 261–273. DOI: 10.1006/jcis.2000.7027.
- [2] K. Skorupski et al. “A fast and accurate implementation of tunable algorithms used for generation of fractal-like aggregate models”. In: *Physica A: Statistical Mechanics and its Applications* 404 (2014), pp. 106–117. DOI: 10.1016/j.physa.2014.02.072.
- [3] L. Ehrl, M. Soos, and M. Lattuada. “Generation and geometrical analysis of dense clusters with variable fractal dimension”. In: *The Journal of Physical Chemistry B* 113 (2009), pp. 10587–10599. DOI: 10.1021/jp903557m.
- [4] H. C. Hamaker. “The London–van der Waals attraction between spherical particles”. In: *Physica* 4 (1937), pp. 1058–1072. DOI: 10.1016/S0031-8914(37)80203-7.
- [5] J. N. Israelachvili. *Intermolecular and Surface Forces*. 3rd ed. Academic Press, 2011.
- [6] P. A. Levin and E. R. Angert. “Small but mighty: Cell size and bacteria”. In: *Cold Spring Harbor Perspectives in Biology* 7 (2015), a019216. DOI: 10.1101/cshperspect.a019216.
- [7] E. Martínez-Salas, J. A. Martín, and M. Vicente. “Relationship of *Escherichia coli* density to growth rate and cell age”. In: *Journal of Bacteriology* 147 (1981), pp. 97–100. DOI: 10.1128/jb.147.1.97-100.1981.

- [8] C. L. Woldringh, J. S. Binnerts, and A. Mans. “Variation in *Escherichia coli* buoyant density measured in Percoll gradients”. In: *Journal of Bacteriology* 148 (1981), pp. 58–63. DOI: 10.1128/jb.148.1.58-63.1981.
- [9] Y. Deng, M. Sun, and J. W. Shaevitz. “Direct measurement of cell wall stress stiffening and turgor pressure in live bacterial cells”. In: *Physical Review Letters* 107 (2011), p. 158101. DOI: 10.1103/PhysRevLett.107.158101.
- [10] J. Lee et al. “Determining the Young’s modulus of the bacterial cell envelope”. In: *ACS Biomaterials Science & Engineering* 10 (2024), pp. 2956–2966. DOI: 10.1021/acsbiomaterials.4c00105.
- [11] G. C. Bushell et al. “On techniques for the measurement of the mass fractal dimension of aggregates”. In: *Advances in Colloid and Interface Science* 95.1 (2002), pp. 1–50. DOI: 10.1016/S0001-8686(00)00078-6.
- [12] J. Theiler. “Estimating fractal dimension”. In: *Journal of the Optical Society of America A* 7.6 (1990), pp. 1055–1073. DOI: 10.1364/JOSAA.7.001055.
- [13] C. M. Sorensen. “Light scattering by fractal aggregates: A review”. In: *Aerosol Science and Technology* 35.2 (2001), pp. 648–687. DOI: 10.1080/02786820117868.
- [14] L. S. Liebovitch and T. Toth. “A fast algorithm to determine fractal dimensions by box counting”. In: *Physics Letters A* 141.8–9 (1989), pp. 386–390. DOI: 10.1016/0375-9601(89)90854-2.
- [15] K. Foroutan-pour, P. Dutilleul, and D. L. Smith. “Advances in the implementation of the box-counting method of fractal dimension estimation”. In: *Applied Mathematics and Computation* 105.2–3 (1999), pp. 195–210. DOI: 10.1016/S0096-3003(98)10096-6.